

A Convolution Neural Networks Scheme for Classification of Motor Imagery EEG based on Wavelet Time-Frequency Image

Hyeon Kyu Lee

Department of Electronics and Communications Engineering
Kwangwoon University, Seoul, Korea
skgusrb12@kw.ac.kr

Young-Seok Choi

Department of Electronics and Communications Engineering
Kwangwoon University, Seoul, Korea
yschoi@kw.ac.kr

Abstract—Motor imagery classification using electroencephalogram (EEG) is increasingly becoming popular in brain-computer interface (BCI) field. In this paper, we present a novel convolution neural networks (CNN) approach for classifying motor imagery electroencephalogram (EEG). For this, we obtained time-frequency representations of EEGs in cases of left and right motor imageries using continuous wavelet transform (CWT). Through experiments using well known public benchmark dataset, it shows that the proposed method provides the improved accuracy compared to the traditional machine learning based classification methods. It confirms the usefulness of the CNN scheme for BCI research field.

Keywords—Electroencephalogram (EEG), Brain-computer interface (BCI), motor imagery, continuous wavelet transform, convolution neural networks (CNN)

I. INTRODUCTION

Brain-computer interface (BCI) is a system that connects the human brain directly to an external device. Among BCI research field, noninvasive BCI has drawn extensive attention, which uses scalp electroencephalogram (EEG) [1]. EEG compared with other brain wave activities is widely used since it requires inexpensive devices and its ease of use. The primary purpose of BCI research is to help a lot of people with severe motor disabilities to control their body using only brain waves [2]. Motor imagery (MI) is a widely used BCI model, which mentally simulates various motor motions, e.g., imaging the motions of limbs. Specifically, it has been known that imaging motor motions yields event related synchronization (ERS) and event related desynchronization (ERD) in the sensorimotor EEGs. The main aim of MI BCI is to control computer and machine using only MI task of brain without actual movement of the subject's body.

Here, we present a novel approach which incorporate a convolution neural networks (CNN) and time-frequency representation. Among the variety of types of MI task, we consider a public benchmark dataset which is comprised of left hand and right hand MI tasks.

Recently, deep learning methods is showing successful performance in various research fields such as computer vision and language recognition. Among various deep learning methods,

the CNN is widely used in image recognition and has a high accuracy. In order to make use of the CNN, we created a 2-dimension (2D) image by applying continuous wavelet transform (CWT) [3]. When applying CWT, we use two motor wavelets, i.e., Morlet and bump wavelets. Also, unlike CNN used in other studies, we propose a CNN model that carries out 1D convolution at the convolution layer.

II. METHOD

A. database

We used dataset 2b from the four BCI competition [4]. This database was proposed by the Graz University of Technology, Austria. The dataset is comprised of motor imagery data about left hand and right hand for nine subjects. Each motor imagery data is extracted by three electrodes (C3, Cz and C4). They were recorded by using bipolar recording with a sampling frequency of 250 Hz. Filter was used bandpass filter between 0.5 Hz and 100 Hz, and a notch filter at 50 Hz. The electrode position Fz used in EEG ground. Each session is made up of three sessions and two class of imagery. In the first two sessions, each trial starts with a fixation cross and an acoustic stimulus. After 3 seconds, a visual arrow orders the left or right MI task. This order is continued for 1.25 seconds. Then the subjects should be performed MI task. This scheme is shown in Fig.1.

After MI for 4 seconds, the subjects have a short break time for least 1.5 seconds. The last three sessions with smiley feedback.

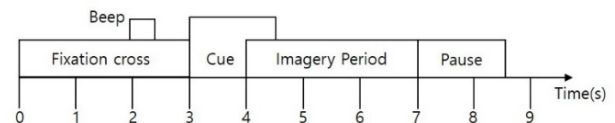


Fig. 1. Scheme of database paradigm

B. Motor Imagery EEG

The public benchmark dataset provides the recording data in cases of left/right hand MI task. In most MI studies, μ and β

This work was supported by Institute for information & communications Technology Promotion(IITP) grant funded by the Korea government(MISP). (No.2017-0-00167, Development of Human Implicit/Explicit Intention Recognition Technologies for Autonomous Human-Things Interaction) and Basic Science Research Program through the National Research Foundation of Korea(NRF) funded by the Ministry of Education (2016R1D1A1B03934014).

bands in sensorimotor EEG were utilized in MI task classification [5]. The power spectral density (PSD) in the μ band (8~13 Hz) decreases during an MI task. This state is called the ERD. On the other hand, the energy of β band (15~38 Hz) in sensorimotor EEG were utilized in MI task classification [5]. The power spectral density (PSD) in the μ band (8~13 Hz) decreases during an MI task. This state is called the ERD. On the other hand, the energy of β band (15~38 Hz) increases, that is called the ERS. According to the left/right hand MI task, the position in the MI cortex of the brain is different. In the MI task for the left hand, ERS and ERD appear in the C3 electrode strongly. On the other hand, ERS and ERD of the MI task for the right hand affect the C4 electrode. We apply the feature of MI task by creating an input image.

C. CNN based learning scheme

We extract EEG segment of 2 second length from each left/right hand MI EEG recording. EEG recordings are sampled as 250 samples per second (Each EEG recording contains 500 samples). Then, each EEG segment is carried out using CWT with two different mother wavelets, i.e., Morlet and Bump wavelets.

Continuous Wavelet Transform (CWT) is used to make a 2D image along with time-frequency property. Since EEG signals possess non-stationarity and non-linearity, CWT has the advantage of keeping signal spectrum information along with time-frequency domain. CWT has mean value of zero and finite wave in time. The parameters of scaling and the shifting in time domain play a critical role in EEG analysis. CWT is defined that the sum over all time of the signal multiplied by scaled, time-shifted versions of the wavelet function ψ [6], [7].

$$CWT(\omega, s) = \frac{1}{\sqrt{|s|}} \int f(t) \psi^* \left(\frac{t-\omega}{s} \right) dt \quad (1)$$

where ψ is the Mother wavelet, ψ^* is the complex conjugate of ψ , ω is scaling and s is shifting parameter.

CWT of each EEG segment yields 331×500 image (Morlet wavelet) and 321×500 image (Bump wavelet). This number denotes the number of time and frequency samples, respectively. And, we extracted power spectrum along with μ and β frequency bands. We extracted EEG time-frequency image corresponding to 6-13 Hz (μ band) and 17-23 Hz (β band). The sizes of extracted time-frequency image are 40×500 for β band and 27×500 for μ band. Then, the size of the time-frequency image is 67×500 for each electrode. In order to reduce the image size, we carry out resizing of the images as follows: First, on the frequency domain side, μ and β bands are resized as 16×50 and 15×500 using the cubic interpolation. By repeating resizing for three electrodes (C3, Cz and C4), a combined time-frequency image is obtained as shown in Fig.2. The size of the time-frequency image of a EEG segment in Fig.2 is 93×500 where $N_{fr} = 93 = N_c \times 31$ and $N_c = 3$.

As a next step, on the time domain side, we extracted data for a 0.5 second period in the smallest mean part. Therefore, the size of an image for a 0.5 second period is 93×125 . The samples

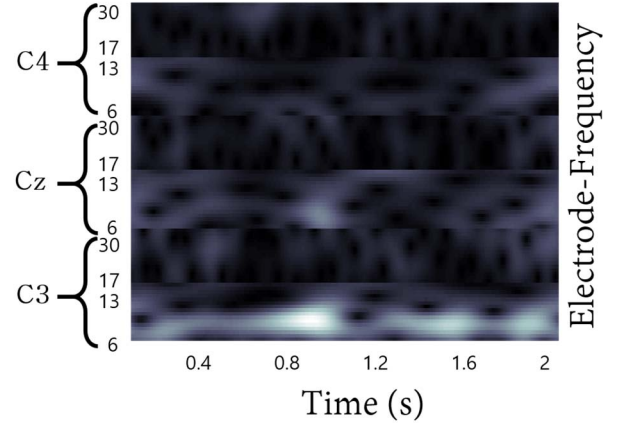
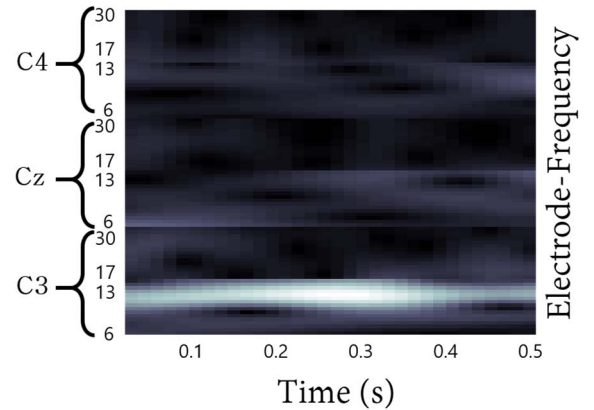
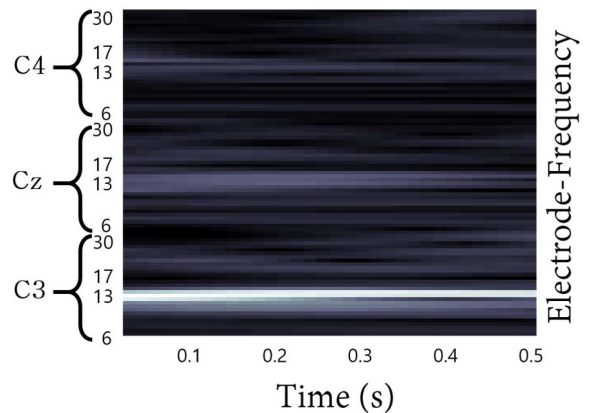


Fig. 2. The resized image form for left hand MI using Morlet WT.



(a)



(b)

Fig.3. The input image form for left hand MI (a) using Morlet wavelet transform, (b) using Bump wavelet transform

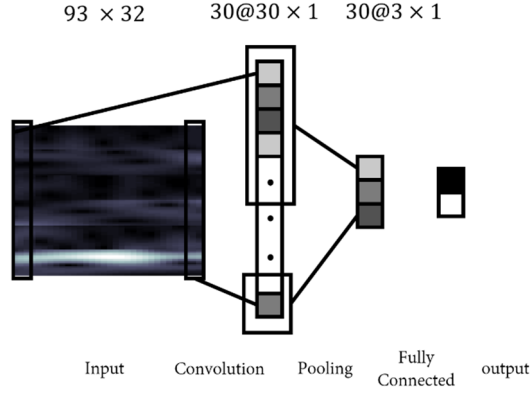


Fig.4. structure of 1D Convolutional Neural Networks

for a 0.5 second reduces 32 samples using cubic interpolation method to fast CNN learning. Finally, these images were obtained $N_{fr} = 93$ and $N_t = 32$. Input image form can confirm in Fig.3.

The response of the motor cortex of brain following MI task can confirm obviously in Fig.3. Along the vertical cortex of the brain, the color in C4 electrode is darker than other electrodes. This mean that The ERD effect in C4 electrode is strong in response to left hand MI task. On the other hand, the color in C3 electrode is lighter than other electrodes. This also means that the ERD effect in C3 electrode is weak in response to left hand MI task. However, the ERS effect in C4 electrode cannot confirm in Fig.3. In the next step, we tried to classify using CNN for the input images.

D. Convolutional Neural Network

CNN is widely used in the field of image processing and computer vision. The structure of CNN consists of convolutional layer, pooling layer, and fully connected layer. While the two-dimensional (2D) convolution filter and subsampling are commonly used in the traditional CNN, here, we conduct 1D convolution at the stage of convolution layer to analyze spectral characteristic over times. Due to more simple CNN model, it yields to faster CNN model compared to previous CNN structures. We utilize 30 filters with size $N_{fr} \times 3$ for convolution stage. The overall model of the CNN is shown in Fig.4. The output of the convolutional layer applies a rectified linear unit (ReLU) function. The use of ReLU function is given as follows as

$$\text{ReLU}(x) = \ln(1 + e^x) \quad (2)$$

The size of the output of convolutional layer through the ReLU function is 30×1 . In addition, the max-pooling layer uses sampling factor of 10, the time-frequency image with a size of 30×1 is subsampled to 3×1 . Then, fully connect layer is performed to classify the output into two classes, i.e., left hand and right hands motor imageries. The parameters of the CNN are learned by the back-propagation algorithm. The error E is calculated by the value for the predetermined label and the output

of CNN classification, it leads to the gradient descent learning scheme as follows as

$$W^k = W^k - \eta \frac{\partial E}{\partial W^k} \quad (3)$$

where η is the learning rate and W^k is the matrix of weights for the filter k .

III. RESULT

The training dataset was used in the study is comprised of 400 trials for each subject. 90% of this data was separated into training set and remaining 10% was separated into test set using 10×10 folds cross-validation. CNN was performed with a batch size of 50 and an epoch of 300. TABLE I shows the means and standard deviations of classification accuracy for each subject using Morlet and Bump wavelet transforms.

TABLE I

Accuracy (%) and standard deviation of MI classification for all subjects

Subjects	Accuracy % (mean $\pm$ standard deviation)	
	Morlet	Bump
1	83.5 ± 7.47	82.5 ± 7.17
2	59.5 ± 9.41	55.75 ± 9.79
3	66.75 ± 7.27	56.75 ± 10.68
4	95.5 ± 4.05	95.5 ± 3.5
5	80 ± 6.12	83.5 ± 7.66
6	75.6 ± 8.72	77 ± 4.97
7	77.25 ± 7.95	76 ± 5.92
8	85.78 ± 4.72	83 ± 5.63
9	86.5 ± 5.79	85.25 ± 4.93
mean	78.93 ± 6.83	77.25 ± 6.69

IV. CONCLUSIONS

Here, we have presented a new usage of CNN for MI EEG patterns, which incorporates 1D CNN structure with the CWT based time-frequency representation. Using two different CWT schemes, it has shown that the integration of CWT based time-frequency images and CNN based classification has achieved the promising performance in MI BCI studies. As future works, we are working on developing and analyzing more effective neural networks schemes on MI BCI researches.

REFERENCE

- [1] B. Z. Allison, E. W. Wolpaw, and J. R. Wolpaw, "Brain-Computer Interfaces System: Progress and Prospects," *Expert Rev. Medical Device*, vol.4, no. 4, pp. 463-474, 2007.

- [2] S. Gao, Y. Wang, X. Gao and B. Hong, "Visual and Auditory Brain-Computer Interfaces," in IEEE Transaction on Biomedical Engineering, vol 61, no 5, May 2014.
- [3] Y. R. Tabor and U. Halici, "A novel deep learning approach for classification of EEG motor imagery signals," in Journal of Neural Engineering, 2017.
- [4] Schlogl A 2003 Outcome of the BCI-Competition 2003 on the Graz Data Set (Berlin : Graz University of Technology)
- [5] G. Pfurtscheller and F. H. Lopes da Silva, "Event-related EEG/MEG synchronization and desynchronization: basic principles," Clinical Neurophysiology, vol. 110, no. 11, pp. 1842–1857, 1999.
- [6] E. Yavuz and O. Aydemir, "Olfaction recognition by EEG analysis using wavelet transform features," in INnovations in Intelligent SysTems and Applications (INISTA), IEEE, pp. 1-4, 2016.
- [7] B. L. Koley and D. Dey, "Detection of Chracteristic Waves of Sleep EEG by Continuous Wavelet Transform," in National Conference on Computing and Communication Systems(NCCCS) IEEE,1-5, India.